TCP CHATROOM USING PYTHON -Vnvs Adithya

SERVER CODE

```
import socket
import threading

# Server Configuration
host = '127.0.0.1'
port = 55555

# Starting Server
server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
server.bind((host, port))
server.listen()

# Lists For Clients and Nicknames
clients = []
nicknames = []

# Broadcast message to all clients
def broadcast(message):
    for client in clients:
        try:
            client.send(message)
        except:
            pass

# Handle messages from a client
def handle(client):
    while True:
        try:
            # Receiving messages
            message = client.recv(1024)
            if message:
```

```
            broadcast(message)
        else:
            remove_client(client)
            break
    except:
        remove_client(client)
        break

# Remove client helper function
def remove_client(client):
    if client in clients:
        index = clients.index(client)
        nickname = nicknames[index]

        clients.remove(client)
        nicknames.remove(nickname)
        client.close()

        broadcast(f"{nickname} left the chat!".encode('ascii'))
        print(f"{nickname} disconnected")

# Accept new clients
def receive():
    print("Server is Listening...")
    while True:
        client, address = server.accept()
        print(f"Connected with {address}")

        client.send("NICK".encode('ascii'))
        nickname = client.recv(1024).decode('ascii')

        nicknames.append(nickname)
        clients.append(client)

        print(f"Nickname is {nickname}")
        broadcast(f"{nickname} joined the chat!".encode('ascii'))
        client.send("Connected to server!".encode('ascii'))

        thread = threading.Thread(target=handle, args=(client,))
        thread.start()

receive()
```

CLIENT CODE

```
import socket
import threading
# Choosing Nickname
nickname = input("Choose your nickname: ")

# Connecting To Server
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect(('127.0.0.1', 55555))
# Listening to Server and Sending Nickname
def receive():
    while True:
        try:
            # Receive Message From Server
            # If 'NICK' Send Nickname
            message = client.recv(1024).decode('ascii')
            if message == 'NICK':
                client.send(nickname.encode('ascii'))
            else:
                print(message)
        except:
            # Close Connection When Error
            print("An error occured!")
            client.close()
            break
        # Sending Messages To Server
def write():
    while True:
        message = '{}: {}'.format(nickname, input(''))
        client.send(message.encode('ascii'))
        # Starting Threads For Listening And Writing
receive_thread = threading.Thread(target=receive)
receive_thread.start()

write_thread = threading.Thread(target=write)
write_thread.start()
```

RESULT

```
Command Prompt - C:\Users\

Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>C:\Users\HP\PycharmProjects\tcpchatroom\.venv\Scripts\python
.exe C:\Users\HP\PycharmProjects\tcpchatroom\server.py
Server is Listening...
Connected with ('127.0.0.1', 55256)
Nickname is Adithya
Connected with ('127.0.0.1', 56936)
Nickname is Harika
```

```
Command Prompt - C:\Users\

Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>C:\Users\HP\PycharmProjects\tcpchatroom\.venv\Scripts\python
.exe C:\Users\HP\PycharmProjects\tcpchatroom\client.py
Choose your nickname: Adithya
Adithya joined the chat!
Connected to server!
Harika joined the chat!
Harika: hey hi adii
hii harry
Adithya: hii harry
```

```
Command Prompt - C:\Users\

Microsoft Windows [Version 10.0.26100.6899]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>C:\Users\HP\PycharmProjects\tcpchatroom\.venv\Scripts\python
.exe C:\Users\HP\PycharmProjects\tcpchatroom\client.py
Choose your nickname: Harika
Harika joined the chat!
Connected to server!
hey hi adii
Harika: hey hi adii
Adithya: hii harry
```